

Python: Recap II

This notebook was generated in **solutions** mode.

Exercise 1: Circle

Exercise Description

Create a Python class called `Circle` that models a geometric circle. The class should have the following features:

1. Attributes:

- radius: A numeric attribute representing the radius of the circle.

2. Methods:

- `__init__(self, radius)`: A constructor method to initialize the radius attribute when a Circle object is created.
- `get_area(self)`: A method that returns the area of the circle (use the formula $\pi \times \text{radius}^2$).
- `get_perimeter(self)`: A method that returns the perimeter (circumference) of the circle (use the formula $2 \times \pi \times \text{radius}$).
- `display_info(self)`: A method that prints the radius, area, and perimeter of the circle.

Your solution

```
import math

class Circle:
    def __init__(self, radius):
        self.radius = radius

    def get_area(self):
        return math.pi * (self.radius ** 2)
```

```
def get_perimeter(self):  
    return 2 * math.pi * self.radius  
  
def display_info(self):  
    print(f"Circle with radius: {self.radius}")  
    print(f"Area: {self.get_area():.2f}")  
    print(f"Perimeter: {self.get_perimeter():.2f}")
```

Test your solution

```
# Test the Circle class  
circle = Circle(5)  
assert circle.radius == 5, "Test failed: Incorrect radius"  
  
# Test area calculation  
area = circle.get_area()  
assert abs(area - 78.54) < 0.1, "Test failed: Incorrect area calculation"  
  
# Test perimeter calculation  
perimeter = circle.get_perimeter()  
assert abs(perimeter - 31.42) < 0.1, "Test failed: Incorrect perimeter calculation"  
  
# Test display_info method exists  
circle.display_info()
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 2: Library Book

Exercise Description

Create a Python class called Book that models a book in a library. The class should have the following features:

1. Attributes:

- `title`: A string representing the title of the book.
- `author`: A string representing the author of the book.
- `isbn`: A string representing the unique ISBN number of the book.
- `is_checked_out`: A boolean attribute indicating if the book is currently checked out (default is `False`).

2. Methods:

- `__init__(self, title, author, isbn)`: A constructor method that initializes the `title`, `author`, `isbn`, and sets `is_checked_out` to `False`.
- `check_out(self)`: Marks the book as checked out by setting `is_checked_out` to `True` and returns `True`. If the book is already checked out, returns `False`.
- `return_book(self)`: Marks the book as returned by setting `is_checked_out` to `False` and returns `True`. If the book is not checked out, returns `False`.
- `get_info(self)`: returns, as a string, the details of the book, including its title, author, ISBN, and availability status.

Your solution

```
class Book:
    def __init__(self, title, author, isbn):
        self.title = title
        self.author = author
```

```

        self.isbn = isbn
        self.is_checked_out = False

    def check_out(self):
        if not self.is_checked_out:
            self.is_checked_out = True
            return True
        else:
            return False

    def return_book(self):
        if self.is_checked_out:
            self.is_checked_out = False
            return True
        else:
            return False

    def get_info(self):
        status = "Available" if not self.is_checked_out else "Checked out"
        return f"Title: {self.title}\nAuthor: {self.author}\nISBN: {self.isbn}\nStatus: {status}"

```

Test your solution

```

# Test the Book class
book = Book("1984", "George Orwell", "123-4567890123")
assert book.title == "1984", "Test failed: Incorrect title"
assert book.author == "George Orwell", "Test failed: Incorrect author"
assert book.isbn == "123-4567890123", "Test failed: Incorrect ISBN"
assert book.is_checked_out == False, "Test failed: Book should be initially available"

# Test check out

```

```
result = book.check_out()
assert result == True, "Test failed: Book should be checked out successfully"
assert book.is_checked_out == True, "Test failed: Book should be marked as checked out"

# Test return
result = book.return_book()
assert result == True, "Test failed: Book should be returned successfully"
assert book.is_checked_out == False, "Test failed: Book should be marked as available"
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 3: Shapes

Exercise Description

Create a base class called Shape with two derived classes, Rectangle and Circle, each with specific attributes. Override the `__eq__` operator in the base class to allow comparison between instances based on their attributes.

Class 1: Shape (Base Class)

- **Attributes:**
 - name: A string representing the name of the shape.
- **Methods:**
 - `__init__(self, name)`: Initializes the name attribute.
 - `__eq__(self, other)`: Returns True if the name and specific attributes of two shapes are identical, and False otherwise.
 - `area(self)`: A placeholder method that should be overridden by subclasses to calculate the area. If called directly, it should generate an error using `raise NotImplementedError`.

Class 2: Rectangle (inherits from Shape)

- **Additional Attributes:**
 - width: The width of the rectangle (a float).
 - height: The height of the rectangle (a float).
- **Methods:**
 - `__init__(self, width, height)`: Initializes the width and height attributes and sets name to "Rectangle".
 - `area(self)`: Returns the area of the rectangle (`width * height`).
 - `__eq__(self, other)`: Checks if two rectangles are equal by comparing width and height values in addition to name.

Class 3: Circle (inherits from Shape)

- **Additional Attributes:**
 - radius: The radius of the circle (a float).
- **Methods:**
 - `__init__(self, radius)`: Initializes the radius attribute and sets name to "Circle".
 - `area(self)`: Returns the area of the circle ($\pi * \text{radius}^2$).
 - `__eq__(self, other)`: Checks if two circles are equal by comparing their radius values in addition to name.

Your solution

```
# Base Class
class Shape:
    def __init__(self, name):
        self.name = name

    def __eq__(self, other):
        if not isinstance(other, Shape):
            return False
        return self.name == other.name

    def area(self):
        raise NotImplementedError("Subclasses should implement the area method")

# Derived Class: Rectangle
class Rectangle(Shape):
    def __init__(self, width, height):
        super().__init__("Rectangle")
```

```

        self.width = float(width)
        self.height = float(height)

    def area(self):
        return self.width * self.height

    def __eq__(self, other):
        if not isinstance(other, Rectangle):
            return False
        return super().__eq__(other) and self.width == other.width and self.height == other.height

    def __str__(self):
        return f"{self.name} (Width: {self.width}, Height: {self.height})"

# Derived Class: Circle
class Circle(Shape):
    def __init__(self, radius):
        super().__init__("Circle")
        self.radius = float(radius)

    def area(self):
        return math.pi * (self.radius ** 2)

    def __eq__(self, other):
        if not isinstance(other, Circle):
            return False
        return super().__eq__(other) and self.radius == other.radius

    def __str__(self):
        return f"{self.name} (Radius: {self.radius})"

```

Test your solution

```
# Test Rectangle initialization
rect = Rectangle(3, 4)
assert rect.width == 3.0, "Rectangle width initialization failed"
assert rect.height == 4.0, "Rectangle height initialization failed"
assert rect.name == "Rectangle", "Rectangle name initialization failed"

# Test Circle initialization
circle = Circle(5)
assert circle.radius == 5.0, "Circle radius initialization failed"
assert circle.name == "Circle", "Circle name initialization failed"

import math

# Test area calculation for Rectangle
rect = Rectangle(3, 4)
assert rect.area() == 12.0, "Rectangle area calculation failed"

# Test area calculation for Circle
circle = Circle(5)
expected_area = math.pi * (5 ** 2)
assert math.isclose(circle.area(), expected_area, rel_tol=1e-9), "Circle area calculation failed"

# Test basic equality between two shapes of the same type
rect1 = Rectangle(3, 4)
rect2 = Rectangle(3, 4)
rect3 = Rectangle(5, 6)

assert rect1 == rect2, "Rectangles with the same dimensions should be equal"
assert rect1 != rect3, "Rectangles with different dimensions should not be equal"
```

```
# Test equality for circles
circle1 = Circle(5)
circle2 = Circle(5)
circle3 = Circle(10)

assert circle1 == circle2, "Circles with the same radius should be equal"
assert circle1 != circle3, "Circles with different radii should not be equal"

# Test inequality between different shape types
assert rect1 != circle1, "Rectangle and Circle should not be equal"
```

Test your solution

```
# Test Rectangle initialization
rect = Rectangle(3, 4)
assert rect.width == 3.0, "Rectangle width initialization failed"
assert rect.height == 4.0, "Rectangle height initialization failed"
assert rect.name == "Rectangle", "Rectangle name initialization failed"

# Test Circle initialization
circle = Circle(5)
assert circle.radius == 5.0, "Circle radius initialization failed"
assert circle.name == "Circle", "Circle name initialization failed"

import math

# Test area calculation for Rectangle
rect = Rectangle(3, 4)
assert rect.area() == 12.0, "Rectangle area calculation failed"
```

```

# Test area calculation for Circle
circle = Circle(5)
expected_area = math.pi * (5 ** 2)
assert math.isclose(circle.area(), expected_area, rel_tol=1e-9), "Circle area calculation failed"

# Test basic equality between two shapes of the same type
rect1 = Rectangle(3, 4)
rect2 = Rectangle(3, 4)
rect3 = Rectangle(5, 6)

assert rect1 == rect2, "Rectangles with the same dimensions should be equal"
assert rect1 != rect3, "Rectangles with different dimensions should not be equal"

# Test equality for circles
circle1 = Circle(5)
circle2 = Circle(5)
circle3 = Circle(10)

assert circle1 == circle2, "Circles with the same radius should be equal"
assert circle1 != circle3, "Circles with different radii should not be equal"

# Test inequality between different shape types
assert rect1 != circle1, "Rectangle and Circle should not be equal"

```

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from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 4: Vectors

Exercise Description

Create two Python classes, `Vector2D` and `Vector3D`, to represent vectors in 2D and 3D space. The `Vector3D` class should inherit from `Vector2D`, adding functionality specific to 3D vectors. Override arithmetic operators to perform vector operations in both classes.

Class 1: Vector2D

- **Attributes:**
 - `x`: The x-coordinate of the vector (a float).
 - `y`: The y-coordinate of the vector (a float).
- **Methods:**
 - `__init__(self, x, y)`: Initializes the `x` and `y` attributes.
 - `__add__(self, other)`: Defines vector addition (+ operator).
 - `__sub__(self, other)`: Defines vector subtraction (- operator).
 - `__mul__(self, scalar)`: Defines scalar multiplication (* operator) where `scalar` is a float or an int.
 - `__str__(self)`: Returns a string representation of the vector such as "(x, y)".

Class 2: Vector3D (inherits from Vector2D)

- **Additional Attributes:**
 - `z`: The z-coordinate of the vector (a float).
- **Methods:**
 - `__init__(self, x, y, z)`: Initializes `x`, `y` (from `Vector2D`), and `z` attributes.
 - `__add__(self, other)`: Overrides addition to handle 3D vector addition.
 - `__sub__(self, other)`: Overrides subtraction to handle 3D vector subtraction.
 - `__mul__(self, scalar)`: Overrides multiplication to handle 3D vector multiplication.
 - `__str__(self)`: Returns a string representation of the 3D vector such as "`(x, y, z)`".

Your solution

```
class Vector2D:
    def __init__(self, x, y):
        self.x = float(x)
        self.y = float(y)

    def __add__(self, other):
        if isinstance(other, Vector2D):
            return Vector2D(self.x + other.x, self.y + other.y)
        raise ValueError("Addition is only supported between Vector2D instances.")

    def __sub__(self, other):
        if isinstance(other, Vector2D):
            return Vector2D(self.x - other.x, self.y - other.y)
```



```

        raise ValueError("Subtraction is only supported between Vector2D instances.")

    def __mul__(self, scalar):
        if isinstance(scalar, (int, float)):
            return Vector2D(self.x * scalar, self.y * scalar)
        raise ValueError("Multiplication is only supported with a scalar.")

    def __str__(self):
        return f"({self.x}, {self.y})"

class Vector3D(Vector2D):
    def __init__(self, x, y, z):
        super().__init__(x, y)
        self.z = float(z)

    def __add__(self, other):
        if isinstance(other, Vector3D):
            return Vector3D(self.x + other.x, self.y + other.y, self.z + other.z)
        raise ValueError("Addition is only supported between Vector3D instances.")

    def __sub__(self, other):
        if isinstance(other, Vector3D):
            return Vector3D(self.x - other.x, self.y - other.y, self.z - other.z)
        raise ValueError("Subtraction is only supported between Vector3D instances.")

    def __mul__(self, scalar):
        if isinstance(scalar, (int, float)):
            return Vector3D(self.x * scalar, self.y * scalar, self.z * scalar)
        raise ValueError("Multiplication is only supported with a scalar.")

```

```
def __str__(self):  
    return f"({self.x}, {self.y}, {self.z})"
```

Test your solution

```
# Test initialization  
v2d = Vector2D(3, 4)  
assert v2d.x == 3.0, "Vector2D initialization failed for x"  
assert v2d.y == 4.0, "Vector2D initialization failed for y"  
  
# Test addition  
v2d_1 = Vector2D(1, 2)  
v2d_2 = Vector2D(3, 4)  
v2d_add = v2d_1 + v2d_2  
assert v2d_add.x == 4.0 and v2d_add.y == 6.0, "Vector2D addition failed"  
  
# Test subtraction  
v2d_sub = v2d_2 - v2d_1  
assert v2d_sub.x == 2.0 and v2d_sub.y == 2.0, "Vector2D subtraction failed"  
  
# Test scalar multiplication  
v2d_mul = v2d_1 * 2  
assert v2d_mul.x == 2.0 and v2d_mul.y == 4.0, "Vector2D scalar multiplication failed"  
  
# Test string representation  
assert str(v2d) == "(3.0, 4.0)", "Vector2D string representation failed"  
  
# Test initialization  
v3d = Vector3D(1, 2, 3)  
assert v3d.x == 1.0, "Vector3D initialization failed for x"  
assert v3d.y == 2.0, "Vector3D initialization failed for y"
```

```

assert v3d.z == 3.0, "Vector3D initialization failed for z"

# Test addition
v3d_1 = Vector3D(1, 2, 3)
v3d_2 = Vector3D(4, 5, 6)
v3d_add = v3d_1 + v3d_2
assert (v3d_add.x, v3d_add.y, v3d_add.z) == (5.0, 7.0, 9.0), "Vector3D addition failed"

# Test subtraction
v3d_sub = v3d_2 - v3d_1
assert (v3d_sub.x, v3d_sub.y, v3d_sub.z) == (3.0, 3.0, 3.0), "Vector3D subtraction failed"

# Test scalar multiplication
v3d_mul = v3d_1 * 2
assert (v3d_mul.x, v3d_mul.y, v3d_mul.z) == (2.0, 4.0, 6.0), "Vector3D scalar multiplication failed"

# Test string representation
assert str(v3d) == "(1.0, 2.0, 3.0)", "Vector3D string representation failed"

```

Test your solution

```

# Test initialization
v2d = Vector2D(3, 4)
assert v2d.x == 3.0, "Vector2D initialization failed for x"
assert v2d.y == 4.0, "Vector2D initialization failed for y"

# Test addition
v2d_1 = Vector2D(1, 2)
v2d_2 = Vector2D(3, 4)
v2d_add = v2d_1 + v2d_2
assert v2d_add.x == 4.0 and v2d_add.y == 6.0, "Vector2D addition failed"

```

```
# Test subtraction
v2d_sub = v2d_2 - v2d_1
assert v2d_sub.x == 2.0 and v2d_sub.y == 2.0, "Vector2D subtraction failed"

# Test scalar multiplication
v2d_mul = v2d_1 * 2
assert v2d_mul.x == 2.0 and v2d_mul.y == 4.0, "Vector2D scalar multiplication failed"

# Test string representation
assert str(v2d) == "(3.0, 4.0)", "Vector2D string representation failed"

# Test initialization
v3d = Vector3D(1, 2, 3)
assert v3d.x == 1.0, "Vector3D initialization failed for x"
assert v3d.y == 2.0, "Vector3D initialization failed for y"
assert v3d.z == 3.0, "Vector3D initialization failed for z"

# Test addition
v3d_1 = Vector3D(1, 2, 3)
v3d_2 = Vector3D(4, 5, 6)
v3d_add = v3d_1 + v3d_2
assert (v3d_add.x, v3d_add.y, v3d_add.z) == (5.0, 7.0, 9.0), "Vector3D addition failed"

# Test subtraction
v3d_sub = v3d_2 - v3d_1
assert (v3d_sub.x, v3d_sub.y, v3d_sub.z) == (3.0, 3.0, 3.0), "Vector3D subtraction failed"

# Test scalar multiplication
v3d_mul = v3d_1 * 2
assert (v3d_mul.x, v3d_mul.y, v3d_mul.z) == (2.0, 4.0, 6.0), "Vector3D scalar multiplication failed"
```

```
# Test string representation
assert str(v3d) == "(1.0, 2.0, 3.0)", "Vector3D string representation failed"
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 5: Books and Library

Exercise Description

Create two Python classes, `Library` and `Book`, to represent a system where a library can manage a collection of books. The classes should have the following features:

Class 1: `Book`

1. **Attributes:**

- `title`: A string representing the title of the book.
- `author`: A string representing the author of the book.
- `isbn`: A string representing the unique ISBN number of the book.
- `is_checked_out`: A boolean attribute indicating if the book is currently checked out (default is `False`).

2. **Methods:**

- `__init__(self, title, author, isbn)`: Initializes the `title`, `author`, `isbn`, and sets `is_checked_out` to `False`.
- `check_out(self)`: Marks the book as checked out by setting `is_checked_out` to `True` and returns `True`. If the book is already checked out, returns `False`.
- `return_book(self)`: Marks the book as returned by setting `is_checked_out` to `False` and returns `True`. If the book is not checked out, returns `False`.
- `get_info(self)`: returns, as a string, the details of the book, including its title, author, ISBN, and availability status.
- `get_title(self)`: returns, as a string, the title of the book.

Class 2: Library

1. Attributes:

- name: A string representing the name of the library.
- books: A list that holds instances of Book objects.

2. Methods:

- `__init__(self, name)`: Initializes the library with a name and an empty list of books.
- `add_book(self, book)`: Adds a Book instance to the library's collection and returns True. If the book is already in the library, returns False.
- `list_books(self)`: returns a list of the titles of all books in the library.
- `list_available_books(self)`: returns a list of the titles of all available books in the library.
- `check_out_book(self, isbn)`: Checks out a book by its ISBN if it is available, returning True. If the book is not available or does not exist, returns False.
- `return_book(self, isbn)`: Returns a book by its ISBN if it is checked out, returning True. If the book is not checked out or does not exist, returns False.

Your solution

```
class Book:
    def __init__(self, title, author, isbn):
        self.title = title
        self.author = author
```

```

        self.isbn = isbn
        self.is_checked_out = False

    def check_out(self):
        if not self.is_checked_out:
            self.is_checked_out = True
            return True
        else:
            return False

    def return_book(self):
        if self.is_checked_out:
            self.is_checked_out = False
            return True
        else:
            return False

    def get_info(self):
        status = "Available" if not self.is_checked_out else "Checked out"
        return f"Title: {self.title}\nAuthor: {self.author}\nISBN: {self.isbn}\nStatus: {status}"

    def get_title(self):
        return self.title

class Library:
    def __init__(self, name):
        self.name = name
        self.books = []

    def add_book(self, book):

```



```
    if not isinstance(book, Book):
        return False
    if book not in self.books:
        self.books.append(book)
        return True
    else:
        return False

def list_books(self):
    books = []
    for book in self.books:
        books.append(book.get_title())
    return books

def list_available_books(self):
    books = []
    for book in self.books:
        if not book.is_checked_out:
            books.append(book.get_title())
    return books

def check_out_book(self, isbn):
    for book in self.books:
        if book.isbn == isbn:
            return book.check_out()
    return False

def return_book(self, isbn):
    for book in self.books:
        if book.isbn == isbn:
```

```
        return book.return_book()
    return False
```

Test your solution

```
# Test the Library and Book classes
library = Library("City Library")
assert library.name == "City Library", "Test failed: Incorrect library name"
assert len(library.books) == 0, "Test failed: Library should start empty"

# Create and add books
book1 = Book("1984", "George Orwell", "123-4567890123")
book2 = Book("To Kill a Mockingbird", "Harper Lee", "456-7890123456")

assert library.add_book(book1) == True, "Test failed: Book should be added successfully"
assert library.add_book(book2) == True, "Test failed: Second book should be added successfully"

# Test listing books
books_list = library.list_books()
assert "1984" in books_list, "Test failed: Book should be in the list"
assert "To Kill a Mockingbird" in books_list, "Test failed: Second book should be in the list"

# Test checking out a book
result = library.check_out_book("123-4567890123")
assert result == True, "Test failed: Book should be checked out successfully"
```

Test your solution

```
# Test the Library and Book classes
library = Library("City Library")
```

```

assert library.name == "City Library", "Test failed: Incorrect library name"
assert len(library.books) == 0, "Test failed: Library should start empty"

# Create and add books
book1 = Book("1984", "George Orwell", "123-4567890123")
book2 = Book("To Kill a Mockingbird", "Harper Lee", "456-7890123456")

assert library.add_book(book1) == True, "Test failed: Book should be added successful
assert library.add_book(book2) == True, "Test failed: Second book should be added suc

# Test listing books
books_list = library.list_books()
assert "1984" in books_list, "Test failed: Book should be in the list"
assert "To Kill a Mockingbird" in books_list, "Test failed: Second book should be in

# Test checking out a book
result = library.check_out_book("123-4567890123")
assert result == True, "Test failed: Book should be checked out successfully"

```

Debug your solution

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```

Exercise 6: Books and Ebooks

Exercise Description

Create a Python class called EBook that inherits from a base class Book. The EBook class will represent a digital version of a book with additional features related to digital media.

Base Class: Book

1. **Attributes:**

- `title`: A string representing the title of the book.
- `author`: A string representing the author of the book.
- `isbn`: A string representing the unique ISBN number of the book.
- `is_checked_out`: A boolean attribute indicating if the book is currently checked out (default is `False`).

2. **Methods:**

- `__init__(self, title, author, isbn)`: Initializes the `title`, `author`, `isbn`, and sets `is_checked_out` to `False`.
- `check_out(self)`: Marks the book as checked out by setting `is_checked_out` to `True` and returns `True`. If the book is already checked out, returns `False`.
- `return_book(self)`: Marks the book as returned by setting `is_checked_out` to `False` and returns `True`. If the book is not checked out, returns `False`.
- `get_info(self)`: Returns a string with the details of the book, including its `title`, `author`, `ISBN`, and `availability status`.

Derived Class: `EBook`

